TABLE 7
Experiment series of the protocol, research question, status and evidence class. M: measured on benchmark data; L: scored against the independent labels J_B; S: simulated evidence or attacker scored against J_B; Y: synthetic registries. [a]Agreement with J_B; human study not run. [b]Observed effects need executed servers. [c]Task success needs E13. [d]64-item set, three models. [e]Needs executed MCP servers. [f]Deterministic repairs only.

ID	Series	RQ	Status	Evid.
E1	Registry characterisation	RQ1	run	M, L
E2	Semantic typing	RQ1	proxy[a]	L
E3	Declared vs. observed effects	RQ2	partial[b]	L
E4	Exposure under policy encodings	RQ2	run	S
E5	Forged and laundered metadata	RQ2	run	S
E6	Imperfect verifier	RQ2	run	S
E7	Trust policy and evidence strength	RQ2	run	S
E8	Single-tool retrieval	RQ3	run	M
E9	Multi-tool set retrieval	RQ3	run	M
E10	Set validity	RQ3	run	L
E11	Capability gap	RQ1, RQ3	partial[c]	M
E12	Model-profile calibration	RQ1	partial[d]	M
E13	End-to-end execution	–	blocked[e]	–
E14	Prompt injection	RQ2, RQ3	offline[e]	L
E15	Cost and scalability	RQ3, RQ4	run	Y
E16	Registry and model evolution	RQ4	run	M
E17	Counterexample-guided refinement	RQ4	partial[f]	M
E18	Mechanism ablation (2^3)	all	run	M, L
E19	LLM role configurations	RQ1	not run	–
E20	Cost weights, utility floor	RQ3	run	M
E21	Experience reuse	RQ4	blocked[e]	–
E22	Statistical validity	all	run	–

8 Experimental Evaluation

This section evaluates the approach, called PaMCP below, against the experimental protocol of the four research questions. Table 7 lists the 22 series of the protocol, the question each answers, whether it was run and the kind of evidence it produces. All results come from a reference implementation of the capability model, Algorithms 1–3 and the certificate. Series that need executed MCP servers (end-to-end task success, E13; observed effects, E3; executed injected actions, E14; experience reuse as in AutoTool [1], E21) or human annotators (E2) could not be run and are reported as such; no claim below rests on them.

8.1 Experimental Setup

Registries and tasks. LiveMCPTool [2] (69 servers, 525 tools) and the tool set of MCPToolBench++ [3] (10 servers, 87 tools) are merged by `server::name`; 43 tools and 3 servers coincide, giving the 76 servers and 569 tools of Table 1. Multi-step tasks are the 95 LiveMCPBench tasks of the public release, 94 of them with a resolvable gold tool. MCP-ToolBench++ provides 1,509 single-call requests; a 60-query development split (ten per category) is used for every tuning decision, and a disjoint evaluation sample of 300 distinct queries (50 per category) is used wherever a method needs a recorded LLM output. E14, E17, E18 and E20 run on 155 tasks, the 95 LiveMCPBench tasks and the 60 development prompts, each under its least-privilege policy (read access plus the scoped writes its goals need) and under a strict read-only policy.

Reference labels. Neither benchmark labels effects. Labels J_A were produced by Claude Sonnet from each tool's name, description and schema under a fixed guide; they provide the typing τ and drive a simulated sandbox verifier bound to $h(t)$. Labels J_B were produced blind by Claude Opus under a separately written guide and are used only for scoring, so that the evidence an experiment simulates never comes from the labels that score it. The two agree on state modification for 558 of the 569 tools ($\kappa = 0.944$) and on private output with $\kappa = 0.893$. Every tool that is not read-only under J_B (120 tools) is also not read-only under J_A (131 tools), so an error-free verifier simulated from J_A can over-exclude but never admit a tool that J_B considers unsafe; verifier errors are injected separately (E5–E7). Both labellings come from one model family and neither is human ground truth.

LLM roles. All LLM outputs were recorded once and are replayed: the analyst's goal contracts (Claude Sonnet; 158 prompts of the scenarios and of the test and development tasks, and the 300 sampled requests), MCP-Zero's model-written requests for 455 tasks and its 76 server summaries, 450 generated multi-tool training queries for COLT, equivalence judgements on 638 candidate pairs for ToolScope's merger, and the calibration answers and self-assessments of three models. The exact model version strings were not exposed to the experiments.

Methods. PaMCP is decomposed into its three mechanisms: the model-conditioned gap (G, Algorithm 1), provenance-sound eligibility (P, Section 5) and set-level reasoning with certification (S: backward closure, minimum-cost cover and certificate, Section 6). Switching S off retrieves one eligible tool per subgoal; switching P off accepts published annotations and lexical cues at face value. The baselines are RAG-MCP [4] (dense top-k over the request), MCP-Zero [5] (recorded requests, server routing, top-1 tool per request), COLT [6] (its collaborative stage as a parameter-free graph propagation over the query–scene–tool graph of the training queries), ToolScope [7] (LLM-confirmed merging into 16 clusters with provider identities and effects kept, dense retrieval over the analyst's sub-queries), CMTF [8] (breadth-first search over typed states with one tool per step), per-subtask retrieval and the full catalogue. All methods share the registry, the contracts and one dense encoder, a TF-IDF and truncated-SVD stand-in (BGE-M3 [9] could not be installed); none of the baselines was tuned. PaMCP uses the parameters of Section 6 with the relevance term $w_{\mathrm{rel}}(1-\hat{s}(e))$ added to Eq. (6) ($w_{\mathrm{rel}} = 10$, selected on the development split).

Metrics and statistics. Exp@k is the share of requests whose top-k list contains a tool that is not read-only under J_B, over the requests whose gold tools are all read-only; Util@k is the share of those requests with a gold tool in the top-k. Group recall is the share of a task's gold groups (alternative tools of one step count once) present in the toolkit. Joint feasibility is decided by an independently written set evaluator that uses only J_B and the analyst's goals: every goal needing the outside world is served (S), every required input is obtainable (C), effects (E) and flows (L) respect the task's policy, evidence meets the policy level (V) and no tool is redundant (M). Paired comparisons use sign-flip permutation tests and bootstrap intervals, with target tools as clusters for request-level metrics, and proportions use Wilson intervals. A confirmatory family of seven comparisons (F1–F7, Table 10), a non-inferiority margin of

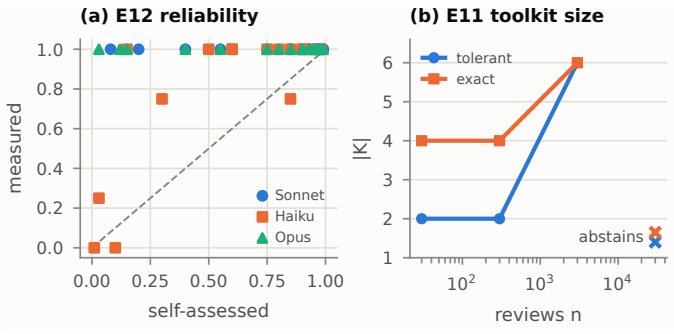


Fig. 8. (a) Self-assessed against measured reliability per class and input size (E12; four items per point). (b) Toolkit size of the review scenario against the number of reviews, Claude Sonnet profile, $\tau = 0.9$ (E11); crosses mark abstentions.

0.05 for recall and Holm correction were fixed before E9, E10, E14 and E18 were run; every other comparison is exploratory. Code, recorded LLM outputs, per-item results and a notebook that re-runs every series on a standard CPU runtime and compares the outcome with the reported results are available.[1]

8.2 Representation and Model Profile (RQ1)

Registry and typing (E1, E2). The merged registry reproduces Table 1. Under J_B, 120 tools (21.1%, 95% CI 17.9–24.6%) are not read-only; 24 of them belong to MCPToolBench++. One of the 44 annotated tools, `opgg-mcp::lol-summoner-renewal`, declares `readOnlyHint=true` but is not read-only. Reachability (S) depends on the typing τ: the LLM-assisted typing agrees with J_B on 78.9% of output kinds ($\kappa = 0.718$, macro-F1 0.704) but only on 35.7% of complete types (kind and domain), whereas a structural typer working from schemas and names alone agrees on 22.8% of kinds ($\kappa = 0.070$) and is never more than 0.7 confident. Lexical or structural typing is therefore not an alternative to an LLM pass, and the type domains are where the two LLM labellings disagree most.

Model profile (E12). Claude Sonnet and Claude Opus answered all 64 calibration items correctly, including exact sums of 640 numbers read from context; Claude Haiku answered 51, failing exact sums from $n = 160$ and exact counts at $n = 640$. Asked beforehand, all three underestimated themselves (mean signed error -0.23 for Sonnet, -0.22 for Haiku and -0.30 for Opus; mean absolute error 0.23, 0.24 and 0.30; Fig. 8a). A self-assessed profile would thus send to tools work that the model performs correctly, which supports measuring ρ_{LLM} rather than asking for it. On the test and development tasks the measured profile overruled 12 of the analyst's 128 native claims (9.4%).

Capability gap (E11). On the review scenario of Fig. 1 the gap grows with the input (Fig. 8b): with tolerant aggregation the toolkit has two tools at $n = 30$ and 300, with 60% of the goals left to the model, and six at $n = 3{,}000$; exact aggregation adds two tools at the smaller sizes. At $n = 30{,}000$ the engine abstains, as it does at $n = 3{,}000$ with $\tau = 0.6$. Without gap analysis it abstains in every setting, because no registry tool serves the scenario's interpretive goals. The three measured profiles give identical toolkits on this scenario. Task success for these toolkits requires execution (E13) and was not measured.

8.3 Trust under Missing, Forged and Imperfect Evidence (RQ2)

Declared effects (E3). Declarations cover 44 tools, one of them wrongly; the other 525 tools, among them 119 that are not read-only, declare nothing. Lexical cues flag 82.5% of the unsafe tools at a precision of 0.54. Observed effects need probe traces of executed servers and were not produced.

Policy encodings (E4). Fig. 9a compares the encodings of a read-only policy. Without a filter, and under MCP's default of displaying annotations without enforcing them, 42.5% of the requests receive an unsafe tool among PaMCP's top five (26.4% for MCP-Zero). Requiring a declared read-only hint keeps 44 tools, one of them unsafe; because the pool is so small, that tool reaches 31.5% of the top-five lists while utility falls to 0.4%. The lexical encoding admits 21 of the 120 unsafe tools (Exp@5 9.9%, Util@5 41.0%). Verified evidence admits none of them and keeps 77.0% of the registry eligible; Util@5 is 46.2% against 49.8% without any filter for PaMCP's retrieval stage, and 72.2% against 76.6% for MCP-Zero. As the verifier probes fewer tools, exposure stays at zero and utility follows the eligible share (34.7%, 26.8% and 16.3% at coverage 0.75, 0.5 and 0.25). The exposure difference to the lexical encoding is confirmatory test F4 ($\Delta = -0.099$, $p_H < 0.001$); by the construction noted in Section 8.1, its zero side reflects the simulation, and the measured content is the utility cost.

Forged and laundered metadata (E5). Forged annotations are what break the closed-world reading: its exposure rises from 31.5% to 57.1% when a tenth of the servers forge their hints and launder their descriptions (56.8% with forgery alone) and to 62.6% at 30% (Fig. 9b). Laundered descriptions barely affect the lexical encoding (8.7–9.9%), because the tool names that carry most lexical cues are kept. Verified evidence does not depend on δ: with a 5% false-accept rate its exposure averages 5.4–5.6% for every $\delta > 0$, at a utility of 46.1–46.7%. Which unsafe tools are falsely accepted varies with the seed, and so does exposure (0.0–14.7% per seed); the single seed run at $\delta = 0$ gives 0.0%.

Imperfect verifier (E6). Exposure of verified evidence grows with the false-accept rate φ, to 1.1%, 5.7%, 9.3% and 15.8% at φ of 1%, 5%, 10% and 20% with $\delta = 0.2$ (Fig. 9c). Errors concentrated on look-alike unsafe tools are not worse (4.4% at 5%, 13.6% at 20%). Verified evidence stays below the lexical encoding (9.2%) up to $\varphi = 5\%$ with random errors and up to 10% with look-alike errors, which quantifies the condition of Section 7 that a verifier must be accurate to help. False rejections cost utility instead: with $\psi = 0.2$, Util@5 falls from 46.2% to 39.1%.

Trust policy (E7). With probe coverage κ varying between tools ($\delta = 0.2$, $\varphi = 5\%$), raising $\ell_{\min}$ from 0.55 or less to 0.65, the level of observed evidence in the implementation, lowers exposure from 5.5% to 3.5% and utility from 45.8% to 34.4%; requiring $\kappa \geq 0.8$ in addition lowers them to 0.6% and 14.3%. Above 0.75 only attested evidence qualifies; exposure then rises to 10.3%, because the few falsely accepted

1. https://github.com/msellamiTN/taas-provenance-discovery, branch `research/pamcp-final-evaluation`, notebook `notebooks/colab/pamcp_evaluation_reviewers.ipynb`.

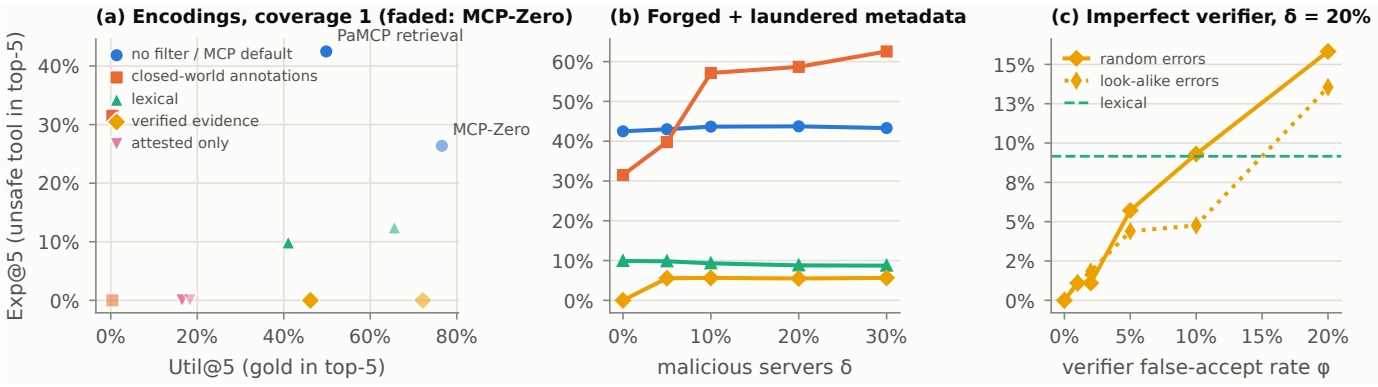


Fig. 9. Eligibility under a read-only policy over 394 requests (273 with read-only gold tools). (a) Exposure against utility of each policy encoding at full verification coverage, for PaMCP's retrieval stage and (faded) MCP-Zero (E4). (b) Exposure as a fraction δ of servers forge their annotations and launder their descriptions; verifier false-accept rate 5%, five seeds for $\delta > 0$, one at $\delta = 0$ (E5). (c) Exposure of verified evidence against the verifier's false-accept rate, with errors at random or concentrated on unsafe tools that resemble safe ones, $\delta = 0.2$ (E6).

attested tools reach many top-five lists of a pool of 25.4% of the registry. Weighting evidence by signer lowers utility (18.4% at 0.65) and raises exposure (6.5% against 3.5%). A stricter threshold therefore does not lower Exp@k by itself; exposure has to be read together with the size of the eligible pool.

8.4 Composition (RQ3)

Single-tool retrieval (E8). Over the MCPToolBench++ tools, retrieval from the analyst's goal contract ranks the gold tool first for 66.0% of the requests, behind MCP-Zero's model-written requests (68.0%) and ahead of raw-text fusion (54.3%), ToolScope and CMTF (57.7%), BM25 (52.7%) and COLT (47.3%) (Fig. 10a). Over the 569-tool registry it falls to 23.0%, the lowest of all methods (MCP-Zero 50.0%, BM25 39.7%, raw-text fusion 38.7%, ToolScope 36.0%, COLT 35.0%, CMTF 33.3%), and the distractor curve shows the loss growing with every block of added tools (Fig. 10b). Confirmatory test F7 thus goes against PaMCP ($\Delta = -0.157$, $p_H < 0.001$). The cause is the structural component of Eq. (5): the analyst's goal text alone ranks better than the raw request (40.0% against 38.7%), but only 50.3% of the gold tools are type-compatible with the analyst's goal output type, so the structural ranking promotes compatible wrong tools (23.0%). Typed retrieval therefore helps only where goal and tool typings agree; with the present typing the structural component would have to be weighted down or applied after relevance, which was not evaluated.

Set retrieval (E9). Table 8 reports the selected sets on the 94 LiveMCPBench tasks. PaMCP exposes the smallest toolkits apart from CMTF (1.47 tools, 199 tokens) and reaches a group recall of 0.169. Methods that expose more tools reach more gold groups: per-subtask retrieval 0.249 with 2.59 tools, RAG-MCP and COLT 0.293 and 0.295 with five tools and over 1,000 tokens, ToolScope 0.361 with 5.20 tools. Non-inferiority to per-subtask retrieval is not established (F1: $\Delta = -0.080$, 95% CI $[-0.166, 0.002]$). At equal size, PaMCP reaches 0.189 against 0.130 for RAG-MCP (F2: $\Delta = 0.060$, $p_H = 0.397$), 0.138 for per-subtask retrieval, 0.084 for COLT and 0.200 for MCP-Zero. A larger candidate budget ($K = 10, 20$) raises recall to 0.174, and backward closure changes it on 3 of the 94 tasks. PaMCP abstains on 7.4% of the tasks (six χ_1 and one χ_5 counterexamples) and certifies the other 92.6%. The analyst's decomposition matters for every method: per-subtask retrieval reaches 0.105 on the raw request, 0.142 with a clause splitter and 0.249 with the analyst's subgoals.

TABLE 8
Set retrieval (E9, open policy) and set validity (E10, each task's least-privilege policy) on the 94 LiveMCPBench tasks. Rec.: group recall; Comp.: share of tasks whose gold groups are all covered; Rec.$_{sm}$: recall with every method truncated to PaMCP's toolkit size (84 tasks with a non-empty toolkit); Joint: joint feasibility under the independent evaluator (PaMCP with K=10, identical for K=20, 40; baselines restricted to eligible tools / as published); Abst.: abstentions. Per-subtask uses the analyst's subgoals. Tokens of the raw tool definitions.

Method	$\lvert K\rvert$	Tok.	Rec.	Comp.%	Rec.$_{sm}$	Joint%	Abst.%
PaMCP (K=5)	1.47	199	0.169	4.3	0.189	67	7
without closure	1.40	195	0.168	4.3	–	–	10
Per-subtask	2.59	346	0.249	9.6	0.138	63/52	0
MCP-Zero	1.76	250	0.227	8.5	0.200	57/55	0
RAG-MCP top-5	5.00	1244	0.293	13.8	0.130	61/49	0
COLT top-5	5.00	1094	0.295	12.8	0.084	60/48	0
ToolScope top-5	5.20	976	0.361	17.0	–	73/60	0
CMTF (path)	0.64	105	0.072	1.1	–	26/23	0

Set validity (E10). Under each task's least-privilege policy, 67.0% of PaMCP's toolkits ($K = 10$; the same for 20 and 40) are jointly feasible for the independent evaluator, 72.4% of those it did not abstain on. The interface, effect, flow and evidence checks pass for 97.9–100%; the failures are goals that the evaluator, which decides from J_B's types rather than τ, does not consider served (S passes for 68.1%, abstentions included). Baselines as published reach 47.9–59.6% and fail the effect check for 4.3–25.5% of the tasks; restricted to eligible tools they gain 2–14 points and reach 57.4–73.4%, ToolScope highest, and CMTF reaches 25.5%. The difference to per-subtask retrieval over eligible tools is not significant (F3: $\Delta = 0.043$, $p_H = 1.0$). Certification raises PaMCP's joint feasibility from 59.6% (NoCert) to 67.0%, through obtainable inputs (C from 88.3% to 97.9%) and minimality (M from 74.5% to 87.2%, against 21.3–41.5% for the top-five retrievers).

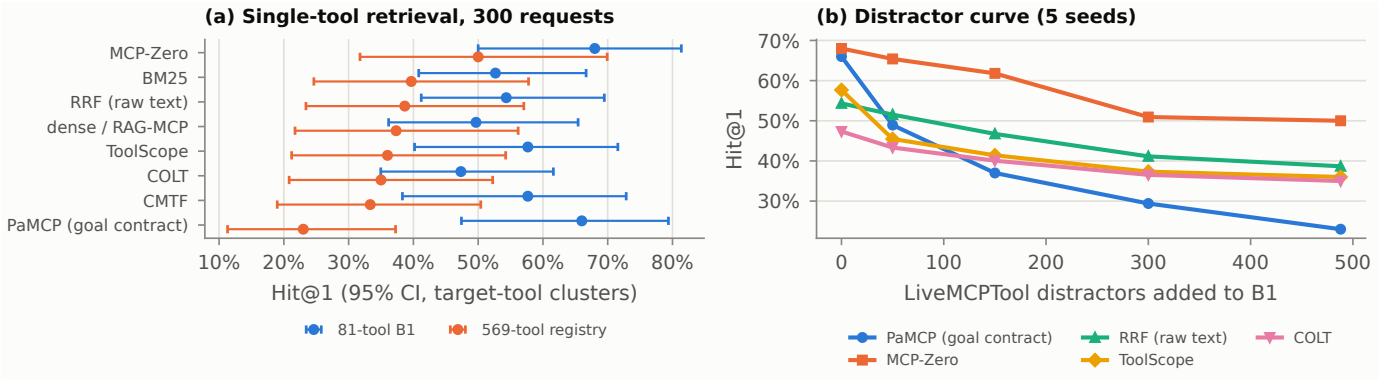


Fig. 10. Single-tool retrieval on the 300-query MCPToolBench++ sample (E8). (a) Hit@1 over the 81 MCPToolBench++ tools (B1) and over the 569-tool registry, with target-tool clustered 95% CIs. (b) Hit@1 as LiveMCPTool tools are added to B1 (five seeds).

TABLE 9
Mechanism ablation (E18) and injection reach (E14) on 155 tasks under each task's least-privilege policy. G: model-conditioned gap; P: provenance-sound eligibility; S: set-level reasoning and certificate. Joint: joint feasibility (independent evaluator); Reach: worst-case share of task × out-of-scope attack pairs the toolkit can realise under J_B (least privilege: 712 pairs; strict read-only: 775). Shapley attribution of the full-vs-none difference (G/P/S): Rec.: −0.017/−0.013/−0.060; Tok.: −55.3/0.8/−66.3; Joint: 0.010/0.026/−0.010; Reach: −0.012/−0.018/−0.022.

Cell	Mech.	$\|K\|$	Tok.	Rec.	Joint%	Reach%	R-O%	Abst.%
Full PaMCP	G P S	1.23	176	0.181	56.1	0.0	0.0	10.3
NoCert	G P –	1.46	229	0.253	56.8	0.1	0.4	0.0
Declared	G – S	1.25	189	0.230	54.8	0.0	0.0	6.5
NoGap	– P S	1.50	193	0.183	55.5	0.0	0.0	20.0
	G – –	1.46	208	0.239	55.5	0.4	3.7	0.0
	– P –	2.09	316	0.295	58.1	0.1	0.4	0.0
	– – S	1.53	221	0.223	52.9	0.1	0.1	16.1
None	– – –	2.08	297	0.270	53.5	1.1	4.9	0.0
RAG-MCP top-5		5.00	1189	0.393	52.3	8.1	14.3	–
Per-subtask		2.08	308	0.321	51.0	2.7	9.4	–
MCP-Zero		1.46	228	0.340	49.0	1.5	8.4	–
ToolScope top-5		5.30	1042	0.461	58.7	7.6	14.7	–
COLT top-5		5.00	1118	0.395	50.3	7.2	12.9	–
CMTF (path)		0.61	114	0.141	23.9	0.7	2.2	–

8.5 Robustness to Prompt Injection and Mechanism Ablation

Prompt injection (E14). For each of 155 tasks and each attack outside the effect classes the task needs, the attack counts as reachable if the exposed toolkit can realise it under J_B. Full PaMCP, NoGap, Declared and every component variant let none of the 712 task–attack pairs through (Table 9; Wilson upper bound 0.5%), and NoCert one, against 8.1% for RAG-MCP, 7.6% for ToolScope, 7.2% for COLT, 2.7% for per-subtask retrieval, 1.5% for MCP-Zero and 0.7% for CMTF as published (F5: $\Delta = -0.027$, $p_H < 0.001$); under a strict read-only policy the retrieval baselines reach 8.4–14.7% and full PaMCP stays at 0 of 775. Restricting the baselines to eligible tools lowers them to at most 1.4% under least privilege, so most of the protection comes from eligibility. The sanitised manifest shows none of the poisoned descriptions when 10% or 30% of the tool descriptions carry an injected instruction, whereas the raw definitions of the same toolkits contain one in 32.6% of the tasks at 30%, and those of the baselines in 17.4% (CMTF) to 86.2% (ToolScope). Of PaMCP's toolkits, 55.5% contain a tool that returns remote content, but none of them also allows an out-of-scope action, against 3.2–22.6% for the baselines. Whether injected actions are executed in practice needs E13's execution environment.

Mechanism ablation (E18). The 2^3 design of Table 9 varies the three mechanisms on the same 155 tasks, registry and scoring. The mechanisms are not additive (G×S interaction on tokens 32.0, $p < 0.001$), so Fig. 11 attributes the full-versus-none difference by Shapley values. Gap analysis and set-level reasoning share the reduction of the context from 297 to 176 tokens (-55.3 and -66.3); eligibility contributes nothing to it. All three remove reachable actions (from 5.2% of the tasks to none). Recall falls from 0.270 to 0.181, mostly through set-level reasoning (-0.060; main effect $p = 0.042$), and no mechanism changes joint feasibility, which stays between 52.9% and 58.1% in every cell. The token saving of the gap alone, Full against NoGap, is not significant after correction (F6: $\Delta = -16.7$, $p_H = 0.269$), while NoGap doubles abstentions (20.0% against 10.3%). Among the components of the full configuration, removing backward closure raises abstentions from 10.3% to 12.3%, propagating flow labels only along derivations raises recall to 0.236 and lowers abstentions to 5.8% at the price of the order-independence of Property 1, removing the relevance term lowers recall to 0.167, and the utility floor and refinement change little.

8.6 Cost, Evolution and Stability (RQ3, RQ4)

Cost and scalability (E15). On the real registry a request takes 49 ms at the median and 191 ms at the 95th percentile on two virtual CPUs; eligibility takes 0.08 ms and the per-goal retrieval, timed separately, 45 ms, against 12 ms for one dense ranking of the request (RAG-MCP). On registries of cloned contracts, which are near-copies of the real tools, the median grows to 58 ms at 1,000 tools, 281 ms at 10,000 and 2.7 s at 100,000 (95th percentiles 0.8 s, 5.7 s and 16.1 s), and retrieval accounts for almost all of it (2.7 s at 100,000, against 206 ms for RAG-MCP). Listing the eligible tools takes 335 ms at 100,000; the bitmap test alone takes 0.024 ms over 10^6 contracts, an index microbenchmark rather than a measurement of the full engine at that size. Building the engine takes 2.2 s, 2.8 s, 30.3 s and 2,224 s: the implementation materialises the pairwise risk relation eagerly, which is quadratic in the

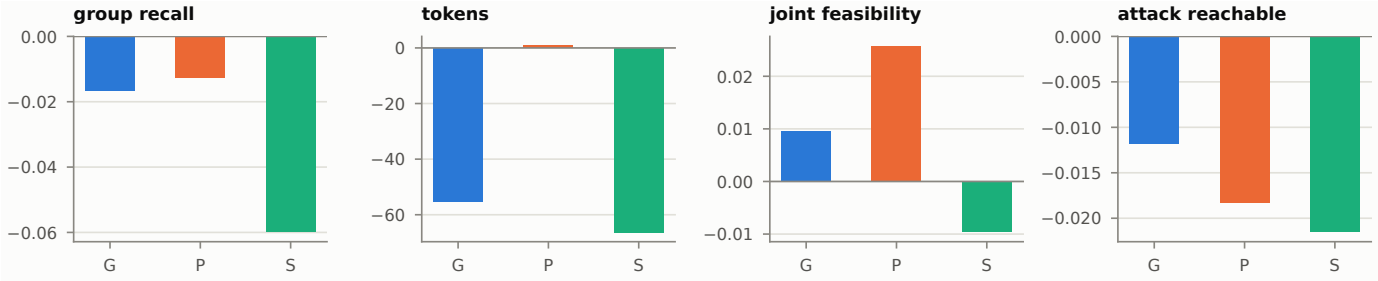


Fig. 11. Shapley attribution of the difference between full PaMCP and the cell without any mechanism to the gap (G), provenance-sound eligibility (P) and set-level reasoning (S), under least-privilege policies (E18): group recall, tokens, joint feasibility and the share of tasks with a reachable out-of-scope action.

number of tools. Compositions use the exact search for 26–28 of the 30 tasks at every size; abstentions (χ_1, and one χ_5) rise from 2 tasks to 9 at 10,000 and 10 at 100,000 tools. On generated cover instances, measured with the objective the search minimises (Eq. (6)), the greedy regime is optimal in 7, 8 and 2 of 10 instances with 1, 3 and 5 goals and on average 0.7%, 2.3% and 14.5% above the optimum (at most 3.0%, 19.8% and 38.1%), while the exact search takes 3 ms, 21 ms and 1.1 s. With 8 and 12 goals the exact search exhausts its 20,000-node budget in all ten instances (after 12.0 s and 18.2 s) and falls back to the greedy regime (27 ms and 69 ms), which carries no optimality guarantee; the gap could not be measured there.

Evolution (E16). Replaying 152 events on the real registry, removing and re-inserting each of the 76 servers, an incremental update takes 13.5 ms at the median (565 ms at the 95th percentile) against 4.2 s for a rebuild. The exact rule of Proposition 3 invalidates 1.1% of the 35 cached toolkits per event, against all of them when the cache is flushed, and after the replay 33 of the 35 recomputed toolkits are identical to the originals. All 20 simulated rug pulls (a description changed after verification) lose their observed evidence, and all regain it when the verified definition returns; a model-version change costs 0.012 ms.

Refinement (E17). Under least-privilege policies 138 of the 155 tasks are certified at once; the deterministic repair operators recover one of the nine χ_1 and none of the eight χ_5 counterexamples (89.0% to 89.7%). When χ_1 is induced by shrinking the candidate budget to $K = 1$, they raise certification from 73.5% to 85.8%, repairing 46.4% of the χ_1, 30% of the χ_5 and all three χ_2 counterexamples, in 1.3 iterations on average. The revenue scenario of Fig. 3 is repaired by the χ_5 split. The assistant role of refinement and the role configurations of E19 were not run.

Sensitivity (E20). Setting β, η or ζ to 0, 0.25 or 1 instead of 0.5, or $u_{\min}$ to 0.4 or 0.6, changes no toolkit, and $u_{\min} = 0.8$ changes 0.7%. The evidence weight γ changes 7.2–20.1% of the toolkits and the token weight δ 2.9–7.2%, with recall between 0.197 and 0.215 against 0.202; only $\gamma = 1$ combined with $\beta \leq 0.25$ lets a tool that J_B forbids into a toolkit (0.014 per toolkit). The price and latency weights cannot act on these registries, which publish neither, so every tool carries the same default quality of service.

Statistical validity (E22). With 300 requests on 66 target tools, intervals that resample target tools are 2.7–3.3 times wider than request-level intervals (33.6 against 10.7 points for raw-text fusion), so request-level intervals would overstate precision. The ranking of the retrieval methods is stable across category-stratified half splits (Kendall's $\tau = 0.68$, same best method in all 200 splits) but less so across tool-disjoint splits ($\tau = 0.43$, same best method in 87.5%). Of the seven confirmatory comparisons (Table 10), two favour PaMCP after correction (F4, F5), one goes against it (F7), non-inferiority of recall is not established (F1), and three are not significant (F2, F3, F6).

TABLE 10
Preregistered confirmatory family (E22): paired sign-flip tests, bootstrap 95% CIs of the mean difference (first method minus second), Holm-adjusted p_H. For F1, p is the one-sided non-inferiority test.

ID	Comparison	n	Δ	95% CI	p	p_H
F1	E9 recall, PaMCP vs. per-subtask (non-inferiority, Δ_0=0.05)	94	−0.080	[−0.166, 0.002]	0.763	1.000
F2	E9 recall at equal size, PaMCP vs. RAG-MCP	84	0.060	[−0.016, 0.137]	0.132	0.397
F3	E10 joint feasibility, PaMCP vs. per-subtask (eligible)	94	0.043	[−0.085, 0.181]	0.641	1.000
F4	E4 Exp@5, verified vs. lexical evidence	273	−0.099	[−0.139, −0.066]	<0.001	<0.001
F5	E14 reachable attacks, PaMCP vs. per-subtask	712	−0.027	[−0.039, −0.015]	<0.001	<0.001
F6	E18 tokens, Full vs. NoGap	155	−16.7	[−34.6, 1.0]	0.067	0.269
F7	E8 Hit@1, goal contract vs. raw-text RRF	300	−0.157	[−0.213, −0.100]	<0.001	<0.001

8.7 Threats to Validity

Construct validity. Exposure, reachability and joint feasibility are measured against LLM labels, not human judgement, and since the labels that simulate evidence include every tool the scoring labels consider unsafe, the zero exposure of verified evidence follows from the simulation. Attackers are non-adaptive and the verifier's only errors are the simulated ones. Group recall and joint feasibility are proxies; task success was not measured. *Internal validity.* One model family provides both labellings, the typing, the analyst and the profiles, so their errors may correlate. Each LLM role was sampled once. The baselines are adaptations: COLT

without encoder fine-tuning, ToolScope without its cross-encoder, MCP-Zero with one round of requests and CMTF with relevance tie-breaking and a bounded search; all share a weak dense encoder and none was tuned. *External validity.* The registries hold 569 tools with sparse annotations and no quality-of-service data; the scalability results use cloned contracts and two virtual CPUs.

REFERENCES

[1] J. Jia and Q. Li, "AutoTool: Efficient tool selection for large language model agents," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.

[2] G. Mo, W. Zhong, J. Chen, Q. Yuan, X. Chen, Y. Lu, H. Lin, B. He, X. Han, and L. Sun, "LiveMCPBench: Can agents navigate an ocean of MCP tools?" *arXiv preprint arXiv:2508.01780*, 2025.

[3] S. Fan, X. Ding, L. Zhang, and L. Mo, "MCPToolBench++: A large scale AI agent model context protocol MCP tool use benchmark," *arXiv preprint arXiv:2508.07575*, 2025.

[4] T. Gan and Q. Sun, "RAG-MCP: Mitigating prompt bloat in LLM tool selection via retrieval-augmented generation," *arXiv preprint arXiv:2505.03275*, 2025.

[5] X. Fei, X. Zheng, and H. Feng, "MCP-Zero: Active tool discovery for autonomous LLM agents," *arXiv preprint arXiv:2506.01056*, 2025.

[6] C. Qu, S. Dai, X. Wei, H. Cai, S. Wang, D. Yin, J. Xu, and J.-R. Wen, "Towards completeness-oriented tool retrieval for large language models," in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*. ACM, 2024, pp. 1930–1940.

[7] M. M. Liu, D. Garcia, F. Parllaku, V. Upadhyay, S. F. A. Shah, and D. Roth, "ToolScope: Enhancing LLM agent tool use through tool merging and context-aware filtering," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026, pp. 34 095–34 119.

[8] R. Suresh Babu and L. Ganesh Iyer, "ToolChoiceConfusion: Causal minimal tool filtering for reliable LLM agents," arXiv:2606.06284, 2026.

[9] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," *arXiv preprint arXiv:2402.03216*, 2024.